MEDIUM · ALGEBRA · ANSWER KEY

Linear inequalities in one or two variables

03

7 answers · Rationale-rich · Teach from doc

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Q1

A

A. $25x + 62y \leq 5{,}440$

Choice A is correct. It's given that a truck can haul a maximum of $5{,}630$ pounds. It's also given that during one trip, the truck will be used to haul a 190-pound piece of equipment as well as several crates. It follows that the truck can haul at most $5{,}630$ - 190, or $5{,}440$, pounds of crates. Since x represents the number of 25-pound crates, the expression $25x$ represents the weight of the 25-pound crates. Since y represents the number of 62-pound crates, $62y$ represents the weight of the 62-pound crates. Therefore, $25x + 62y$ represents the total weight of the crates the truck can haul. Since the truck can haul at most $5{,}440$ pounds of crates, the total weight of the crates must be less than or equal to 5,440 pounds, or $25x + 62y \leq 5{,}440$.

Choice B is incorrect. This represents the possible combinations of the number of 25-pound crates, x, and the number of 62-pound crates, y, the truck can haul during one trip if it can haul a minimum, not a maximum, of 5,630 pounds.

Choice C is incorrect. This represents the possible combinations of the number of 62-pound crates, x, and the number of 25-pound crates, y, the truck can haul during one trip if only crates are being hauled.

Choice D is incorrect. This represents the possible combinations of the number of 62-pound crates, x, and the number of 25-pound crates, y, the truck can haul during one trip if it can haul a minimum, not a maximum, weight of $5{,}630$ pounds and only crates are being hauled.

Q2 D

D.

x	y
442	0
441	-2
440	-4

Choice D is correct. All the tables in the choices have the same three values of x, 440, 441, and 442, so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting 440 for x in the given inequality yields 2440 - y > 883, or 880 - y > 883. Subtracting 880 from both sides of this inequality yields $-y > 3$. Dividing both sides of this inequality by -1 yields $y < -3$. Therefore, when $x = 440$, the corresponding value of y must be less than -3. Substituting 441 for x in the given inequality yields 2441 - y > 883, or 882 - y > 883. Subtracting 882 from both sides of this inequality yields $-y > 1$. Dividing both sides of this inequality by -1 yields $y < -1$. Therefore, when $x = 441$, the corresponding value of y must be less than -1. Substituting 442 for x in the given inequality yields 2442 - y > 883, or 884 - y > 883. Subtracting 884 from both sides of this inequality yields $-y > -1$. Dividing both sides of this inequality by -1 yields $y < 1$. Therefore, when $x = 442$, the corresponding value of y must be less than 1. For the table in choice D, when $x = 440$, the corresponding value of y is -4, which is less than -3; when $x = 441$, the corresponding value of y is -2, which is less than -1; when $x = 442$, the corresponding value of y is 0, which is less than 1. Therefore, the table in choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. When $x = 440$, the corresponding value of y in this table is 0, which isn't less than -3.

Choice B is incorrect. When $x = 440$, the corresponding value of y in this table is 0, which isn't less than -3.

Choice C is incorrect. When $x = 440$, the corresponding value of y in this table is -2, which isn't less than -3.

Q3

B

B. $7216 \leq x \leq 7716$

Choice B is correct. It's given that the model estimates that whales from the genus *Eschrichtius* travel 72 to 77 miles in the ocean each day during their migration. If one of these whales travels 72 miles each day for 16 days, then the whale travels 7216 miles total. If one of these whales travels 77 miles each day for 16 days, then the whale travels 7716 miles total. Therefore, the model estimates that in 16 days of its migration, a whale from the genus *Eschrichtius* could travel at least 7216 and at most 7716 miles total. Thus, the inequality $7216 \leq x \leq 7716$ represents the estimated total number of miles, x, a whale from the genus *Eschrichtius* could travel in 16 days of its migration.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q4

A

A.

x	y
19	18
20	19
21	20

Choice A is correct. The inequality $y < x$ indicates that for any solution to the given system of inequalities, the value of x must be greater than the corresponding value of y. The inequality $x < 22$ indicates that for any solution to the given system of inequalities, the value of x must be less than 22. Of the given choices, only choice A contains values of x that are each greater than the corresponding value of y and less than 22. Therefore, for choice A, all the values of x and their corresponding values of y are solutions to the given system of inequalities.

Choice B is incorrect. The values in this table aren't solutions to the inequality $y < x$.

Choice C is incorrect. The values in this table aren't solutions to the inequality $x < 22$.

Choice D is incorrect. The values in this table aren't solutions to the inequality $y < x$ or the inequality $x < 22$.

Q5

A

A. 34

Choice A is correct. It's given that the truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than 4,600 pounds. If the trailer has a weight of 500 pounds and each box weighs 120 pounds, the expression $500 + 120b$, where b is the number of boxes, gives the combined weight of the trailer and the boxes. Since the combined weight must be no more than 4,600 pounds, the possible numbers of boxes the truck can tow are given by the inequality $500 + 120b \leq 4{,}600$. Subtracting 500 from both sides of this inequality yields $120b \leq 4{,}100$. Dividing both sides of this inequality by 120 yields $b \leq \frac{205}{6}$, or b is less than or equal to approximately 34.17. Since the number of boxes, b, must be a whole number, the maximum number of boxes the truck can tow is the greatest whole number less than 34.17, which is 34.

Choice B is incorrect. Towing the trailer and 35 boxes would yield a combined weight of $4{,}700$ pounds, which is greater than $4{,}600$ pounds.

Choice C is incorrect. Towing the trailer and 38 boxes would yield a combined weight of $5{,}060$ pounds, which is greater than $4{,}600$ pounds.

Choice D is incorrect. Towing the trailer and 39 boxes would yield a combined weight of $5{,}180$ pounds, which is greater than $4{,}600$ pounds.

Q6 C

C. 5

Choice C is correct. Let a equal the number of 120-pound packages, and let b equal the number of 100-pound packages. It's given that the total weight of the packages can be at most 1,100 pounds: the inequality $120a + 100b \leq 1{,}100$ represents this situation. It's also given that the helicopter must carry at least 10 packages: the inequality $a + b \geq 10$ represents this situation. Values of a and b that satisfy these two inequalities represent the allowable numbers of 120-pound packages and 100-pound packages the helicopter can transport. To maximize the number of 120-pound packages, a, in the helicopter, the number of 100-pound packages, b, in the helicopter needs to be minimized. Expressing b in terms of a in the second inequality yields $b \geq 10 - a$, so the minimum value of b is equal to $10 - a$. Substituting $10 - a$ for b in the first inequality results in $120a + 100(10 - a) \leq 1{,}100$. Using the distributive property to rewrite this inequality yields $120a + 1{,}000 - 100a \leq 1{,}100$, or $20a + 1{,}000 \leq 1{,}100$. Subtracting 1,000 from both sides of this inequality yields $20a \leq 100$. Dividing both sides of this inequality by 20 results in $a \leq 5$. This means that the maximum number of 120-pound packages that the helicopter can carry per trip is 5.

Choices A, B, and D are incorrect and may result from incorrectly creating or solving the system of inequalities.

Q7

A

A. -9

Choice A is correct. It's given that the point $x, 53$ is a solution to the given system of inequalities in the xy-plane. This means that the coordinates of the point, when substituted for the variables x and y, make both of the inequalities in the system true. Substituting 53 for y in the inequality $y > 14$ yields $53 > 14$, which is true. Substituting 53 for y in the inequality $4x + y < 18$ yields $4x + 53 < 18$. Subtracting 53 from both sides of this inequality yields $4x < -35$. Dividing both sides of this inequality by 4 yields $x < -8.75$. Therefore, x must be a value less than -8.75. Of the given choices, only -9 is less than -8.75.

Choice B is incorrect. Substituting -5 for x and 53 for y in the inequality $4x + y < 18$ yields $4\text{-}5 + 53 < 18$, or $33 < 18$, which is not true.

Choice C is incorrect. Substituting 5 for x and 53 for y in the inequality $4x + y < 18$ yields $45 + 53 < 18$, or $73 < 18$, which is not true.

Choice D is incorrect. Substituting 9 for x and 53 for y in the inequality $4x + y < 18$ yields $49 + 53 < 18$, or $89 < 18$, which is not true.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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